

Network Topologies

A Complete Student Guide for Cyber Security

Bus | Ring | Star | Mesh | Hybrid | Tree

Introduction

What is Network Topology?

Definition

A **Network Topology** is the arrangement or layout of how devices are connected to each other in a network. It is like a map of your network it shows which devices are connected, how they are connected, and the path that data takes to travel from one device to another.

There are two ways to look at topology:

- **Physical Topology** — The actual physical layout. Where are the cables? Where are the devices placed in the room or building?
- **Logical Topology** — How data actually flows through the network, regardless of the physical setup. The data path might be different from what the physical wiring looks like.

Simple Definition

"Network Topology is the shape or structure of a network it tells us how devices are connected and how data moves between them."

The Six Main Types of Network Topology

Each topology has its own shape, its own strengths, and its own weaknesses all of which are covered in detail in this book:

1. **Bus** — All devices share one single central cable (the backbone).
2. **Ring** — Devices are connected in a closed circular loop.
3. **Star** — All devices connect individually to one central switch or hub.

4. **Mesh** — Every device is directly connected to every other device.
5. **Hybrid** — A combination of two or more different topology types.
6. **Tree** — A branching, hierarchical structure built on a main backbone.

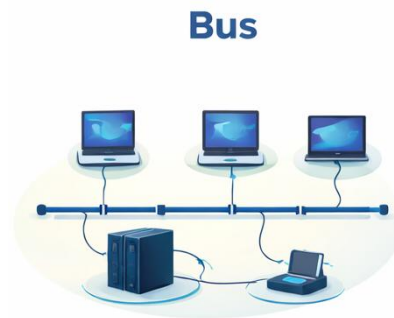
Bus Topology

What is Bus Topology?

In a Bus Topology, all devices are connected to a single central cable called the bus or backbone. Every device shares the same cable. When one device sends data, that data travels along the entire cable and every device receives it but only the device it was meant for will actually read and accept it.

Analogy Imagine a school hallway (the bus cable). Every student (device) stands on the same hallway. When a teacher shouts a student's name, everyone hears it, but only that one student responds.

Diagram



Advantages & Disadvantages

Advantages	Disadvantages
+ Very cheap to set up — needs very little cable	- If the main cable breaks, the entire network goes down
+ Simple and easy to install	- As more devices join, the network gets slower
+ Works well for small networks	- Hard to find which part of the cable is broken
+ Easy to extend by adding more cable	- Low security every device sees all data on the cable
	- Not suitable for large or busy networks

Ring Topology

What is Ring Topology?

In a Ring Topology, every device is connected to exactly two other devices forming a closed circular loop, like a ring. Data travels in one direction around the ring, passing through each device one by one, until it reaches its destination.

Analogy Think of a relay race. The baton (data) is passed from one runner to the next, going around the track (ring), until it reaches the correct runner who takes it.

Diagram



Advantages & Disadvantages

Advantages	Disadvantages
+ Data flows in an orderly way — no data collisions	- If one device or cable fails, the whole network goes down
+ Good performance under heavy traffic	- Slower than star topology for small networks
+ Equal access for all devices on the network	- Adding or removing devices disrupts the whole network
+ Easy to identify which device is causing a problem	- Troubleshooting can be difficult — hard to isolate faults

Star Topology

What is Star Topology?

In a Star Topology, every device is connected individually to a central device usually a switch or hub. No device is directly connected to another device. All data must pass through the central switch to reach its destination.

This is the most widely used topology in modern networks. Your home Wi-Fi router and school computer lab most likely use a star topology.

Analogy Think of an airport hub. All flights (data) must pass through the main hub (switch) before reaching their destination city (device). Planes do not fly directly between small cities.

Diagram



Advantages & Disadvantages

Advantages	Disadvantages
+ If one device fails, the rest of the network keeps working	- If the central switch fails, the entire network stops
+ Easy to add or remove devices without disruption	- Requires more cable than bus topology
+ Easy to troubleshoot faults are easy to isolate	- The switch is a single point of failure
+ Better performance than bus or ring for most tasks	- Cost increases as more devices are added (more ports needed)
+ Most common and well-supported topology today	

Mesh Topology

What is Mesh Topology?

In a Mesh Topology, every single device is connected directly to every other device on the network. This means there are multiple paths between any two devices. If one path is broken or slow, data automatically takes another path.

Analogy Think of a spider web. Every thread connects to multiple other threads. If you break one thread, the web still holds together because data can travel through other threads to reach its destination.

Diagram



Advantages & Disadvantages

Advantages	Disadvantages
+ Extremely reliable no single point of failure	- Very expensive needs a huge amount of cable
+ If one connection fails, data uses another route	- Complex to set up and manage
+ Very high security dedicated links between devices	- Not practical for large networks with many devices
+ Easy to detect and fix faults	- Maintenance is difficult and costly
+ Can handle heavy traffic well	

Hybrid Topology

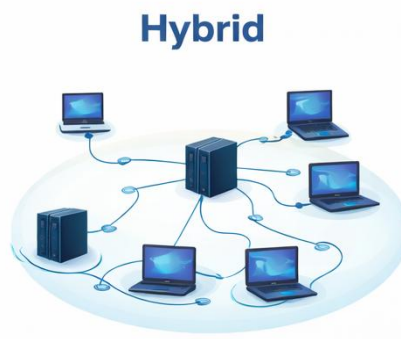
What is Hybrid Topology?

A Hybrid Topology is a combination of two or more different topology types connected together. For example, a company might use a Star topology inside each department and connect all departments together using a Bus or Mesh topology. The result is a hybrid.

Hybrid topology is the most flexible and scalable of all topologies. Large organizations, universities, and enterprise networks almost always use some form of hybrid topology because they can mix and match the best features of each type.

Analogy Think of a city's transport system. Inside your neighbourhood you walk (bus topology), you take a local bus to the nearest metro station (star topology), and then the metro connects multiple stations across the city (mesh topology). It is all connected but uses different methods in different areas.

Diagram



Advantages & Disadvantages

Advantages	Disadvantages
+ Extremely flexible combine the best of multiple topologies	- Complex to design, set up, and manage
+ Highly scalable easy to expand by adding new topology sections	- Expensive combines costs of multiple topologies
+ Reliable faults in one section do not affect others	- Troubleshooting is harder due to mixed topology types
+ Can be optimized for different areas and needs	- Requires skilled IT professionals to maintain
+ Used by most real-world enterprise networks	

Tree Topology

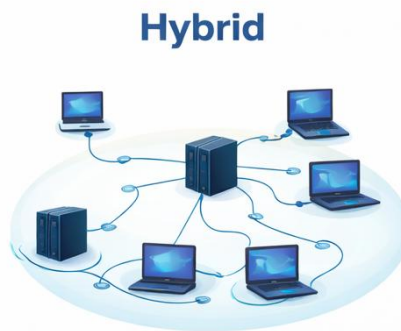
What is Tree Topology?

A Tree Topology (also called a Hierarchical Topology) is shaped like a tree there is one root device at the top, branches spreading out from it, and leaves at the end of each branch. It is essentially a combination of Bus and Star topologies.

There is a main central cable (the trunk) with multiple star-configured networks branching off it. This makes it easy to manage large networks in a structured, organized way. Schools and universities commonly use this topology.

Analogy Think of an actual tree. The roots and trunk (main cable/root switch) support the branches (secondary switches), which support the leaves (end devices like computers and printers). Everything flows through the trunk.

Diagram



Advantages & Disadvantages

Advantages	Disadvantages
+ Easy to manage and expand — just add new branches	- If the root device fails, the entire network goes down
+ Hierarchical structure makes it easy to understand	- Heavy dependence on the main trunk cable
+ A fault in one branch does not affect other branches	- Can be expensive for very large deployments
+ Good for large organizations like schools or companies	- More complex than a simple star topology
+ Easy to isolate and fix problems in specific branches	

Final Comparison

Use this table to quickly compare all six topologies side by side. This is very useful for exam revision!

Feature	Bus	Ring	Star	Mesh	Hybrid	Tree
Cost	Low	Medium	Medium	Very High	High	Medium
Installation	Easy	Medium	Easy	Complex	Complex	Medium
Speed	Slow	Fast	Fast	Very Fast	High	Medium
Reliability	Low	Medium	Medium	Very High	High	Medium
Security	Low	Medium	Medium	High	High	Medium
Scalability	Hard	Hard	Easy	Hard	Easy	Easy
Best For	Old labs	Tokens	Homes/Offices	Military	Enterprises	Schools

Key Takeaway for Cyber Security

The topology of a network directly affects how easy or hard it is to secure. A Mesh topology is the most secure because it has no single point of failure and dedicated links. A Bus topology is the least secure because all devices share the same cable and can see each other's data. As a cyber security professional, always ask: what is the topology? because your defense strategy depends on it.